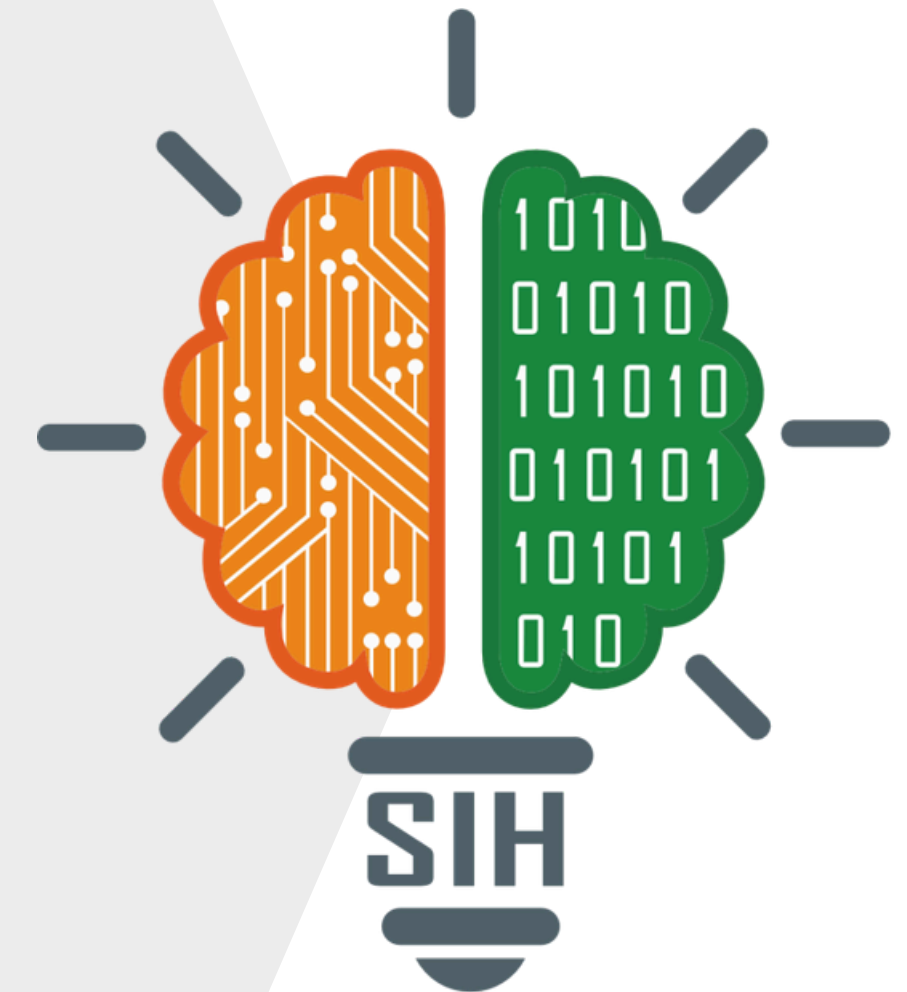


TITLE PAGE

- **Problem Statement ID – SIH26074**
- **Problem Statement Title-** Downscaling of Weather Forecast from Block Level to Panchayat Level for Agro-Meteorological Advisory Services
- **Theme-**
- **PS Category-** SOFTWARE
- **Team ID-**
- **Team Name (Registered on portal) - Team Braket 3.1.0**



THE PROBLEM

Block-level forecasts are **too coarse** for Panchayat-level agricultural decisions.



Same block, different rainfall within Panchayats.



Farmers need location-specific, short-term information.

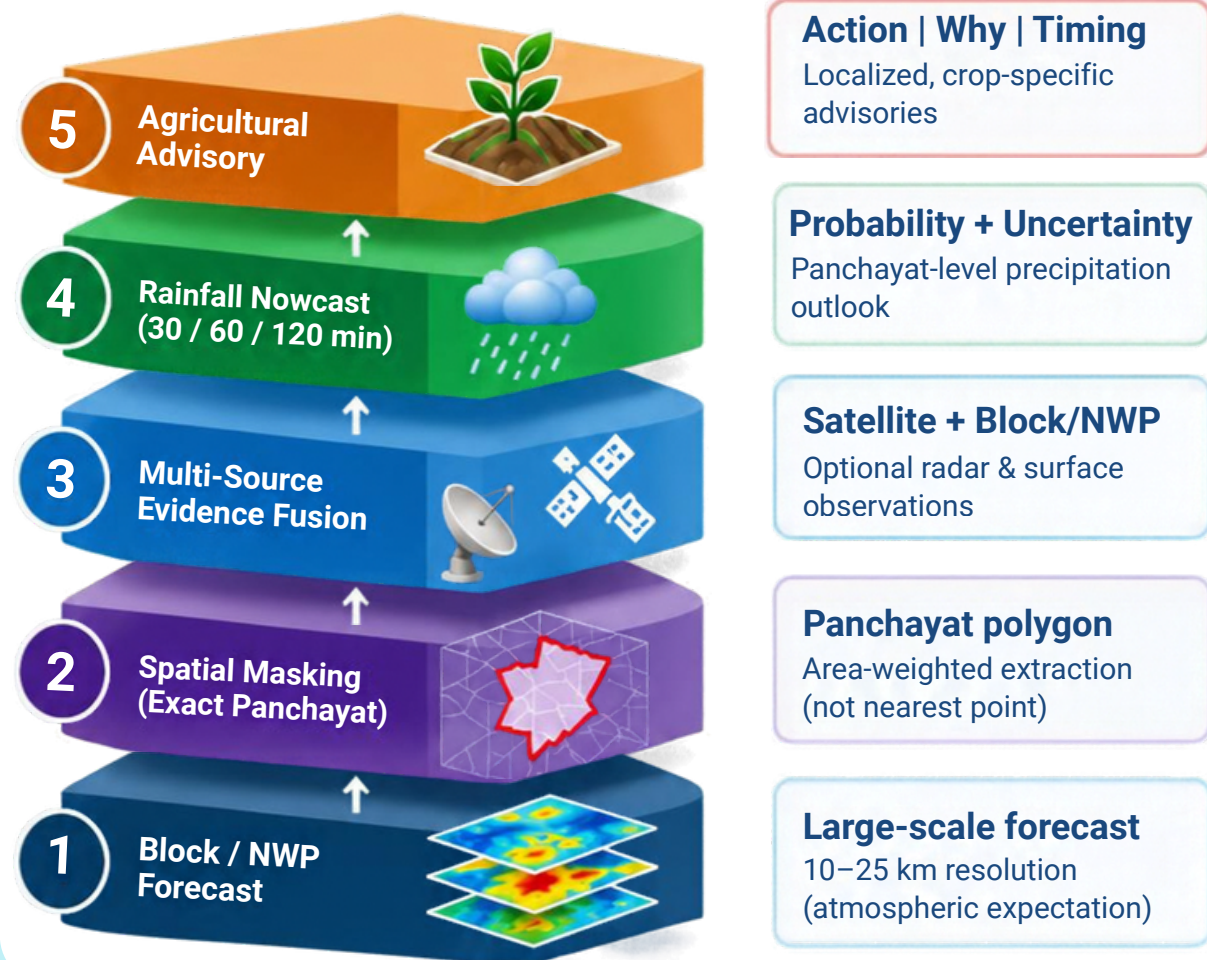


Localized rain events are missed.



Leads to suboptimal farm decisions.

OUR SOLUTION



INNOVATION & UNIQUENESS



Exact Panchayat-level downscaling

Uses real Panchayat polygon with spatial masking, not nearest point.



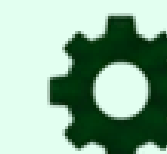
Multi-source evidence fusion

Combines block/NWP, satellite, optional radar and observations.



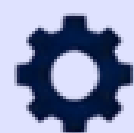
Short-horizon nowcasting

Provides 30/ 60 / 120 minute localized precipitation outlook.



Seamless advisory integration

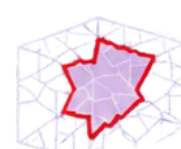
Localized evidence fed into existing agro-meteorological advisory engine.



HOW IT WORKS



User Location (Web / Mobile)



Panchayat Resolution



Satellite + Block/WNP Data



Rainfall Nowcast (30/60/120 min)

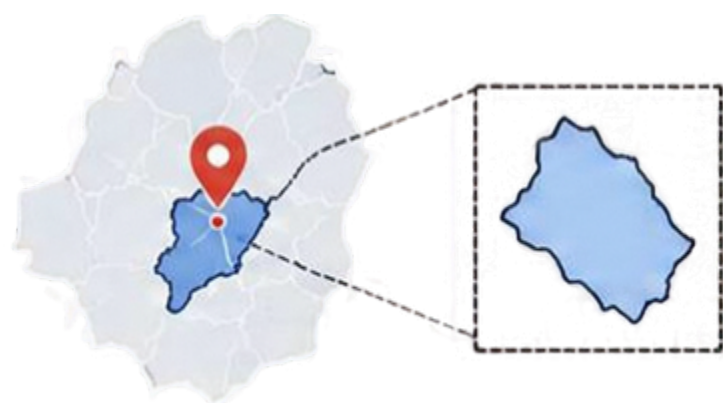


Agricultural Advisory

VIDEO LINK

PROTOTYPE LINK

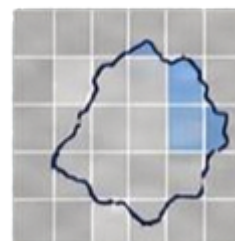
REPORT LINK

1 PANCHAYAT RESOLUTION*Identify exact Panchayat location*

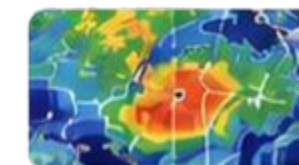
- Point-in-polygon matching
- Exact Panchayat ID
- ON_BOUNDARY if ambiguous

2 BLOCK / NWP BASELINE*Large-scale weather context***Open-Meteo
(Current)****IMD Adapter
(When Authorized)**

- Block-level temperature, precipitation and other parameters
- Used as baseline (existing model retained)

3 SPATIAL MASKING*Map data to Panchayat area***Source Grid (NWP
/ Satellite)****Panchayat Masked
Values**

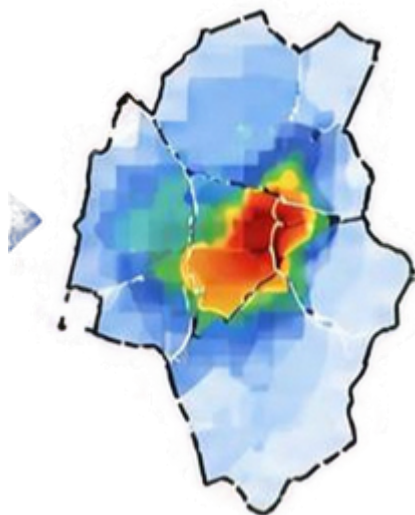
- Polygon grid intersection
- Fractional area weighing
- Preserves native source resolution

4 LOCAL OBSERVATIONS*Localized spatial evidence***INSAT-DRT TIR
(Satellite)****Satellite Precipitation
Estimates****Optional Radar
(when available)****Surface Observations
(optional)**

- Timestamp + freshness + provenance

5 PRECIPITATION NOWCAST*Localized short-horizon forecast*

-  NWP Baseline
-  Sattelite Observations
-  (Operational) Radar
-  (Operational) Surface Data

30 / 60 / 120 min
rainfall outlookProbability of
precipitation

Estimated amount



Confidence / uncertainty

6 ADVISORY INTEGRATION*Convert forecast into actionable advice***What to do?**

Crop & farming recommendation

**Why?**

Weather-based reason

**When?**

Optimal Timing

**Keep Baseline & Localized
context separately****7 WEB & MOBILE DELIVERY***Easy access for farmers & officials*

- React Web + Flutter Mobile
- Panchayat identity & localized forecast
- Confidance & Source taste

TECHNOLOGY STACK**Geospatial**

- Python
- Shapely (STRtree)
- Panchayat Polygons (LGD)

Weather Data

- Open-Meteo
- IMD adapter
- NWP (block-level)

Satellite Data

- INSAT-DRT TIR (Georeferenced grids)
- Precipitation estimates

Backend (APIs)

- Panchayat boundary service
- Spatial masking
- Observation processing
- Precipitation nowcast
- Advisory integration

Frontend

- React (Web)
- Flutter (Mobile)

Governance

- Provenance & freshness
- Confidence scores
- Fail-closed states
- Live / Demo separation



FEASIBILITY

Can we build and deploy it with existing data, tools and systems?



DATA SOURCES READY

NWP + Satellite (INSAT-3DR) + Optional Radar



GEOSPATIAL INFRASTRUCTURE

Exact Panchayat polygons and spatial masking



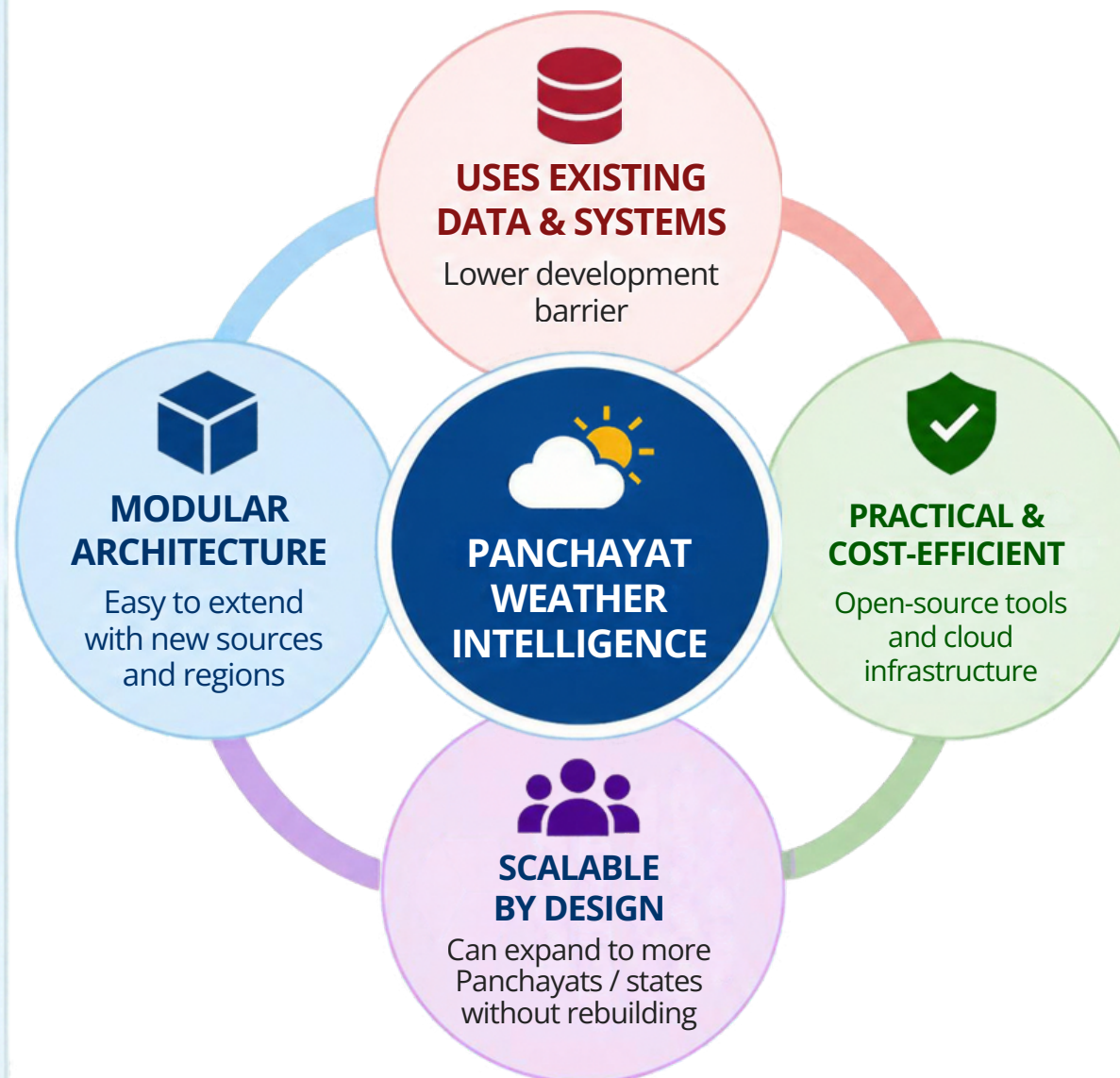
TECHNICALLY IMPLEMENTED

11 stages completed (modular and verifiable)



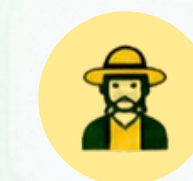
DEPLOYMENT READY

Works on existing React (web) & Flutter (mobile) stack



VIABILITY / IMPACT

Can it create real value and be sustained long-term?



FOR FARMERS

More localized and timely weather evidence for better agricultural decisions



FOR GOVERNMENT

Supports Panchayat-level agro-advisory services and data-driven planning



FOR AGRICULTURAL SYSTEMS

Directly integrates with existing crop/risk advisories for actionable recommendations



NATIONAL SCALE POTENTIAL

Architecture can extend across Panchayats, regions and additional data sources (radar, AWS, more satellites)

FROM DATA TO IMPACT



NWP Block Forecast



Panchayat Boundary



Satellite Evidence (+ optional Radar)



Localized Nowcast (30/60/120 min)



Agro Advisory (Crop / Risk Rules)



Farmer Action



KEY CHALLENGES & MITIGATION



Limited radar coverage
Use satellite + NWP, with confidence levels



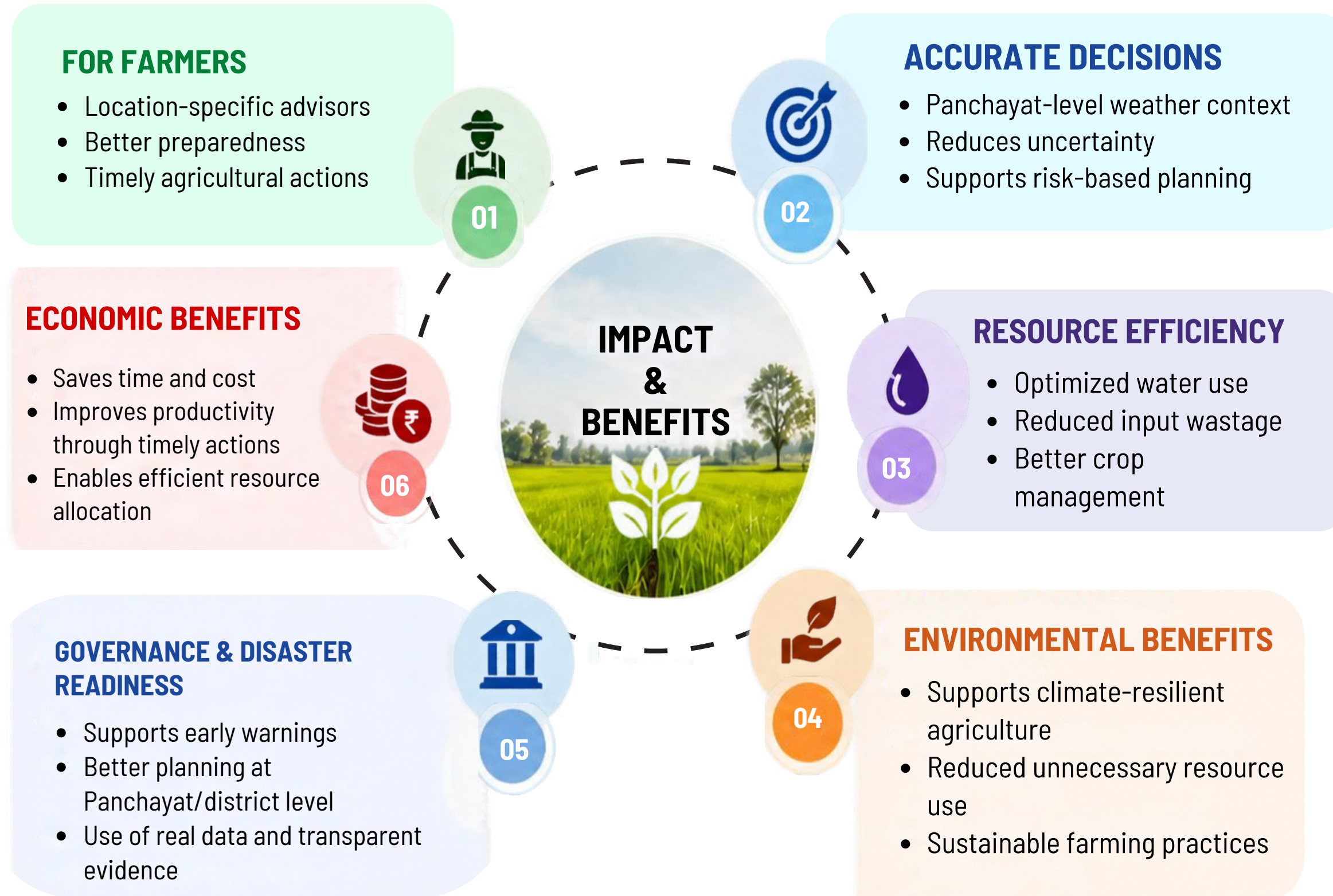
Missing or stale data
Fail-closed / unavailable state



Panchayat boundary ambiguity
Return ON, BOUNDARY, no guessing



Source resolution is coarse
Use spatial masking, preserve native resolution



SAME BLOCK, DIFFERENT PANCHAYAT-LEVEL INSIGHTS

Panchayat A (Rameswar)



84.8%

Rain Probability
(next 30 min)

Higher chance of rain –
suggests rain-sensitive actions

Panchayat B (Jansa)



25.4%

Rain Probability
(next 30 min)

Lower chance of rain –
continue normal farm activities



Both Panchayats are in the same block (Arajine, Varanas) with the same IMD-GFS baseline, but localized evidence differs.

OVERALL IMPACT



**More relevant
advisories at
Panchayat level**



**Timely actions
reduce risk &
uncertainty**



**Supports sustainable
& resilient agriculture**

01

DATA SOURCES & RESEARCH



- Block-level NWP forecast (IMD-GFS)
- Satellite data (INSAT-3DR TIR)
- Optional radar & surface observations
- Authorized Panchayat boundary data
- Relevant research on spatial downscaling and nowcasting

03

RESEARCH CHALLENGES



- Spatial variability within the same block
- Coarse resolution of source data
- Panchayat boundary ambiguity
- Limited and intermittent radar coverage
- Validation at Panchayat scale
- Ensuring data freshness and reliability

05

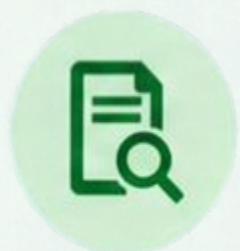
VALIDATION & RESULTS



- Independent Panchayat-scale validation (eastern Uttar Pradesh)
- Real satellite data (INSAT-3DR)
- 30/60/120 min rainfall nowcast
- Pilot results show clear Panchayat-level differences within the same block
- Scientific status: LIMITED VALIDATION

02

BASELINE & EXISTING METHODS



- Block-level NWP forecast as baseline
- Traditional spatial interpolation methods
- Satellite-based precipitation estimates
- Existing agricultural advisory systems
- Review of nowcasting and downscaling approaches

04

OUR APPROACH



- Exact Panchayat polygon + spatial masking
- Multi-source evidence fusion
- Two-stage short-horizon precipitation nowcast (30/60/120 min)
- Integration with existing advisory engine
- Web (React) + Mobile (Flutter) deployment

06

FUTURE SCOPE & REFERENCES



- Extend validation to more Panchayats/regions
- Incorporate additional satellite/radar products
- Higher-resolution (local nowcasting methods)
- Apply to more crops and advisors
- Further research on AI/ML for precipitation downscaling

KEY REFERENCES



Datasets
[Link](#)



Existing Methods
[Link](#)



Research Gaps
[Link](#)



Future Scope
[Link](#)



Experimental Results
[Link](#)